PAGE 3 · HERO LOGICAL ARCHITECTURE

How PeX Resolves a Payment Exception

From exception event to governed resolution and final business outcome.

PAYMENT HUBS (ACI · Finastra · FIS)
Inbound Exception Feeds

CORE BANKING (DDA Ledgers · Accounts)
Posting Breaks & Balances

CLEARING RAILS (SWIFT · Fedwire · CHAPS)
Protocol Rejections & Returns

PeX GOVERNED RESOLUTION ENGINE (LOGICAL ARCHITECTURE)

1. INGEST ➔ NORMALIZE ➔ CLASSIFY
Canonical PaymentExceptionCase + 41-Profile Map

2. EVIDENCE & CONTEXT CORROBORATION
Read-Only ICDs: DDA · Sanctions · KYC · SWIFTRef

3. AI MODEL GATEWAY (ReasoningResult)
Advisory Hypothesis & Confidence Scoring

4. DETERMINISTIC RULES ENGINE
100% Scheme Validation — Rules Override AI

5. POLICY & SANCTIONS RE-SCREENING
Watchlist & Travel Rule Compliance Check

6. HUMAN REVIEW & MAKER-CHECKER
Hard $0 SoD Limit · Dual-Auth for ≥$250k

7. CONTROLLED EGRESS GATEWAY
Generates Schema-Compliant Repair ICD Contract

8. CLEARING DISPATCH (DISPATCHED)
Secure Transport ACK from Central Clearing Rail

9. FINAL SETTLEMENT (SETTLED)
pacs.002 ACCP or Internal DDA Ledger Posting

CROSS-CUTTING GOVERNANCE: Policy Rules | Maker-Checker ($0 SoD) | Dual-Auth (≥$250k) | SHA-256 Merkle Provenance | WORM 7-Yr Audit

CORE ARCHITECTURAL MANTRA: AI recommends. Controls validate. Humans authorize. Systems execute.

LIFECYCLE: EVENT ➔ INTELLIGENCE ➔ DETERMINISTIC CONTROL ➔ HUMAN AUTHORIZATION ➔ EXECUTION ➔ FINAL SETTLEMENT ➔ MERKLE AUDIT